

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and trained **Deep Learning networks** for object classification and scene understanding, iterating on model architectures to improve accuracy on automotive datasets.
- Selected and implemented appropriate **data representation methods** and **dataset sampling strategies** for training computer vision models, ensuring alignment with target automotive use cases.
- Performed **statistical analysis** and fine-tuning of deep learning models using test results, identifying performance bottlenecks and optimizing hyperparameters for better generalization.
- Implemented **Python scripts** to automate data preprocessing and augmentation pipelines, handling large volumes of image data for model training and validation.
- Collaborated with data engineers to ensure optimal **data acquisition** and maintain data integrity across evolving datasets for deep learning applications.
- Containerized deep learning inference services using **Docker** and deployed them to **AWS EC2** instances for testing and integration, managing dependencies and environment configurations.
- Contributed to code reviews and **agile ceremonies**, ensuring adherence to software development best practices within the computer vision team.

SKILLS

Deep Learning – Keras, PyTorch, scikit-learn, TensorFlow

Computer Vision – OpenCV, 3D object detection, Pedestrian classification, Lane detection, Structure-from-motion

Programming – Python, English

Data Handling – Data acquisition, Data pipelines, Data integrity, Statistical analysis, Data modeling

Software Engineering – Linux, Version control, Software architecture, Embedded platforms, Agile

Cloud & Containerization – AWS EC2, Docker

PROJECTS

Deep Learning Model for Vehicle Detection

- Developed and trained a **Convolutional Neural Network (CNN)** using **PyTorch** on a dataset of ~150,000 automotive images to detect and classify vehicles.
- Implemented custom data augmentation techniques in **Python** to increase model robustness against varying lighting and weather conditions, improving detection accuracy by **18%** on challenging edge cases.
- Evaluated model performance using metrics such as **precision** and **recall**, and fine-tuned network architecture and hyperparameters to achieve a target detection accuracy of **0.92**.

Image Preprocessing Pipeline for Sensor Fusion

- Engineered an automated **Python** pipeline to ingest and preprocess image data from multiple automotive sensors, standardizing formats and resolutions for downstream analysis.
- Integrated **OpenCV** for image manipulation tasks including resizing, color correction, and noise reduction, processing approximately **5,000 images** per hour.
- Developed shell scripts for efficient file management and batch processing on a **Linux** environment, reducing manual intervention by **60%**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)